

Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1 - \lambda_c) \delta \mathbf{x}[\mathbf{k}] - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k]f_T[k]$$

d -step measurement delay:

$$\delta \mathbf{x}[\mathbf{k}] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta x_D[k-i]$$

Ideal control effort (with d -step measurement delay):

$$\begin{aligned} \bar{\bar{f}}_{dx}[k] = a_x^{-1}[k] \Big\{ & \underbrace{\left[\Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \\ & - \underbrace{\left[\delta x_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_D[k-i] \right]}_{\delta x_D^d[k]} \Big\} \end{aligned}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\left\{ a_x[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i]f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] [f_{dx}[k] - \bar{\bar{f}}_{dx}[k]] - \delta x_D^d[k]$$

$$a_x[k] \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] - \delta x_D^d[k]$$

$$a_x[k] f_{dx}[k] = (1 + e_{a_x}[k] \hat{a}_x^{-1}[k]) \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[(\delta x[k-d] + n_x[k]) - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &= e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= a_{x_{det}}[k+1] + a_{x_{ram}}[k+1] = \underbrace{\{a_{x_{det}}[k] + a_{x_{ram}}[k]\}}_{a_x[k]} + \Delta a_x[k] + \underbrace{\{a_{x_{ram}}[k+1] - a_{x_{ram}}[k]\}}_{\Delta a_{ram}[k]} \\ &= a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k] \end{aligned}$$

$$\Delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a'_x[k] \Delta h_d[k] \text{ (from } p_d), \quad \Delta a_{ram}[k] = a'_x[k] \Delta x_{ram}[k]$$

$$\Delta h_d[k] = h_d[k+1] - h_d[k] \text{ (desired-trajectory height increment).}$$

Gain dynamics — $a_{x_{ram}}$ is first-order autoregressive (AR(1))

Assumption ($x = x_{det} + x_{ram}$, perfect tracking $x_{det} = x_d$): $\delta x = x_d - x = -x_{ram}$ (zero-mean), $x_{ram} = -\delta x$.

$$a_{x_{ram}}[k+1] = -a'_x \delta x[k+1] = -a'_x (\lambda_c \delta x[k] - \varepsilon[k]) = \lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]$$

$$\begin{aligned} a_x[k+1] &= (a_{x_{det}}[k] + \Delta a_x[k]) + (\lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]) \\ &= (a_{x_{det}}[k] + \Delta a_x[k]) + \lambda_c (a_x[k] - a_{x_{det}}[k]) + a'_x \varepsilon[k] \\ &= \underbrace{\lambda_c a_x[k]}_{\mathbf{F}_e(4,4)=\lambda_c} + \underbrace{(1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k]}_{\text{known input}} + \underbrace{a'_x \varepsilon[k]}_{q_4[k]} \end{aligned}$$

$\mathbf{F}_e(4,4) = \lambda_c$

Parameter estimator (predict reverts to $a_{x_{det}}$): $\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k]$

Parameter estimation error: $e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$

Estimation error dynamics: $e_{a_x}[k+1] = \lambda_c e_{a_x}[k] + a'_x \varepsilon[k]$

$$a_x[k-i] - \hat{a}_x[k-i] = e_{a_x}[k-i] \stackrel{(\dagger)}{\approx} e_{a_x}[k] - \sum_{j=1}^i \Delta a_{ram}[k-j]$$

($\dagger$) feedforward form $e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}$ is exact; the AR(1) predictor gives $e_{a_x}[k+1] = \lambda_c e_{a_x} + a'_x \varepsilon$, altering the expansion by $O((1-\lambda_c)e_{a_x})$ which enters only $\delta x_{\delta af}^d$ ($\ll \delta x_T^d$ in ε).

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &\approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1-\lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ &\quad - \underbrace{\left\{ (1-\lambda_c) \sum_{i=1}^d (d+1-i) f_{dx}[k-i] \Delta a_{ram}[k-i] \right\}}_{\delta x_{\delta af}^d[k]} \\ &\quad + \delta x_D^d[k] + (1-\lambda_c) n_x[k] \end{aligned}$$

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x_D^d[k] - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta af}^d[k] + (1-\lambda_c) n_x[k] \right\}}_{\varepsilon[k]}$$

State equation

($\delta x_D^d \equiv 0$: disturbance not modelled)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = \lambda_c a_x[k] + (1-\lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + a'_x \varepsilon[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \Delta a_x[k-i] \end{bmatrix} \text{ (known)}, \quad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

Past-error expansion under AR(1) (†)

Two predictors, two e_{a_x} recursions (predict-only):

$$\text{feedforward: } e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}[k] \Rightarrow e_{a_x}[k-i] = e_{a_x}[k] - \sum_{j=1}^i \Delta a_{ram}[k-j] \quad (\text{exact})$$

$$\text{AR(1): } e_{a_x}[k+1] = \lambda_c e_{a_x}[k] + a'_x \varepsilon[k] \Rightarrow e_{a_x}[k-i] = \lambda_c^{-i} (e_{a_x}[k] - \sum_{j=1}^i \lambda_c^{j-1} a'_x \varepsilon[k-j])$$

Difference ($\Delta_i := e_{a_x}^{\text{AR}(1)}[k-i] - e_{a_x}^{\text{ff}}[k-i]$):

$$\Delta_i = \underbrace{(\lambda_c^{-i} - 1) e_{a_x}[k]}_{\text{reverting}} - \lambda_c^{-i} \sum_{j=1}^i \lambda_c^{j-1} a'_x \varepsilon[k-j] + \sum_{j=1}^i \Delta a_{ram}[k-j]$$

Leading order (small i , $\lambda_c^{-i} \approx 1 + i(1 - \lambda_c)$):

$$(\lambda_c^{-i} - 1) e_{a_x}[k] \approx i(1 - \lambda_c) e_{a_x}[k] = O((1 - \lambda_c) e_{a_x})$$

$\Rightarrow$ the feedforward expansion errs by $O((1 - \lambda_c) e_{a_x})$, entering ε only through $\delta x_{\delta af}^d$ ($\ll \delta x_T^d$); numerically Q_{33}/R_{22} with $\text{Var}(\Delta a_{ram})$ vs q_4 leaves $\hat{a}_x$ unchanged.

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \left(\hat{a}_x[k] - \sum_{i=1}^d \Delta a_x[k-i] \right) \\ &= \left(a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + r_2[k] \right) - \hat{a}_x[k] + \sum_{i=1}^d \Delta a_x[k-i] \\ &= e_{a_x}[k] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{aligned}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k] e_{\delta x_1}[k] - \ell_{12}[k] e_{a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k] e_{\delta x_1}[k] - \ell_{22}[k] e_{a_x}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k] e_{\delta x_1}[k] - \ell_{32}[k] e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{a_x}[k+1] = \lambda_c e_{a_x}[k] - \ell_{41}[k] e_{\delta x_1}[k] - \ell_{42}[k] e_{a_x}[k] + \underbrace{a'_x \varepsilon[k]}_{q_4[k]}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & \lambda_c \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k - i]$$

$$\mathbf{e}[k + 1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$

Process / measurement noise variances

Assumption ($x = x_{det} + x_{ram}$, perfect tracking): $x_{ram} = -\delta x$ (zero-mean), $a'_x := \frac{da_x}{dh} = -\frac{a_x K_h}{R}$, $K_h = \frac{1}{c} \frac{dc}{dh}$ (wall sensitivity).

Closed-loop tracking variance (thermal-driven, $\hat{a}_x = a_x$):

$$\text{Var}(x_{ram}) = \text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2, \quad C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}, \quad \sigma_{\delta h}^2 := 4k_B T a_x$$

(i) **True increment** $\Delta a_{ram} = a'_x \Delta x_{ram}$ ($\rho_1 = \text{Cov}(x_{ram}[k+1], x_{ram}[k]) / \text{Var}(x_{ram})$, lag-1 autocorr):

$$\text{Var}(\Delta x_{ram}) = 2(1 - \rho_1) \text{Var}(x_{ram}), \quad \rho_1 = \lambda_c + \frac{2(1 - \lambda_c)}{C_{\delta x}} \Rightarrow 1 - \rho_1 = \frac{1}{(1 + \lambda_c) C_{\delta x}}$$

$$\boxed{\text{Var}(\Delta a_{ram}) = a_x'^2 \cdot \underbrace{2(1 - \rho_1) C_{\delta x}}_{= 2/(1 + \lambda_c)} \sigma_{\delta h}^2 = \frac{2}{1 + \lambda_c} a_x'^2 \sigma_{\delta h}^2 \quad (\rightarrow Q_{33}^{\text{randgain}}, R_{22}^{\text{delay}})}$$

(ii) **AR(1) white driver** $w_4 = a'_x \varepsilon$ ($\text{Var}(\varepsilon) = (1 - \lambda_c^2) \text{Var}(\delta x)$ since δx is AR(1)):

$$\boxed{q_4 = a_x'^2 \text{Var}(\varepsilon) = (1 - \lambda_c^2) C_{\delta x} a_x'^2 \sigma_{\delta h}^2 = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2 \quad (\rightarrow Q_{44})}$$

Self-consistency (AR(1) steady-state variance = true gain variance):

$$\frac{q_4}{1 - \lambda_c^2} = C_{\delta x} a_x'^2 \sigma_{\delta h}^2 = \text{Var}(a_{x_{ram}}) \quad \checkmark$$

Impl.: K_h , $\sigma_{\delta h}^2 = 4k_B T a_\perp$ evaluated at $\bar{h}_{\text{meas}}(p_m)$; gain $a_x \rightarrow \hat{a}_x$ (estimate); reversion target $a_{x_{det}}$ from p_d .

Original random-walk Q_{44} — the mismatch

Original model (RW, $\mathbf{F}_e(4,4) = 1$): predict mean and Q_{44} are mutually consistent,

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k], \quad Q_{44} = \text{Var}(\Delta a_{ram}).$$

True increment (from $a_{x_{ram}}[k+1] = \lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]$):

$$\Delta a_{ram}[k] = a_{x_{ram}}[k+1] - a_{x_{ram}}[k] = \underbrace{(\lambda_c - 1) a_{x_{ram}}[k]}_{\text{reversion (spring)}} + \underbrace{a'_x \varepsilon[k]}_{\text{white}}$$

Mismatch: RW puts the *whole* Δa_{ram} into Q_{44} — it treats the deterministic reversion $(\lambda_c - 1)a_{x_{ram}}$ as white noise; $\mathbf{F}_e(4,4) = 1$ keeps no spring, so the modelled gain variance is unbounded while the true gain is bounded:

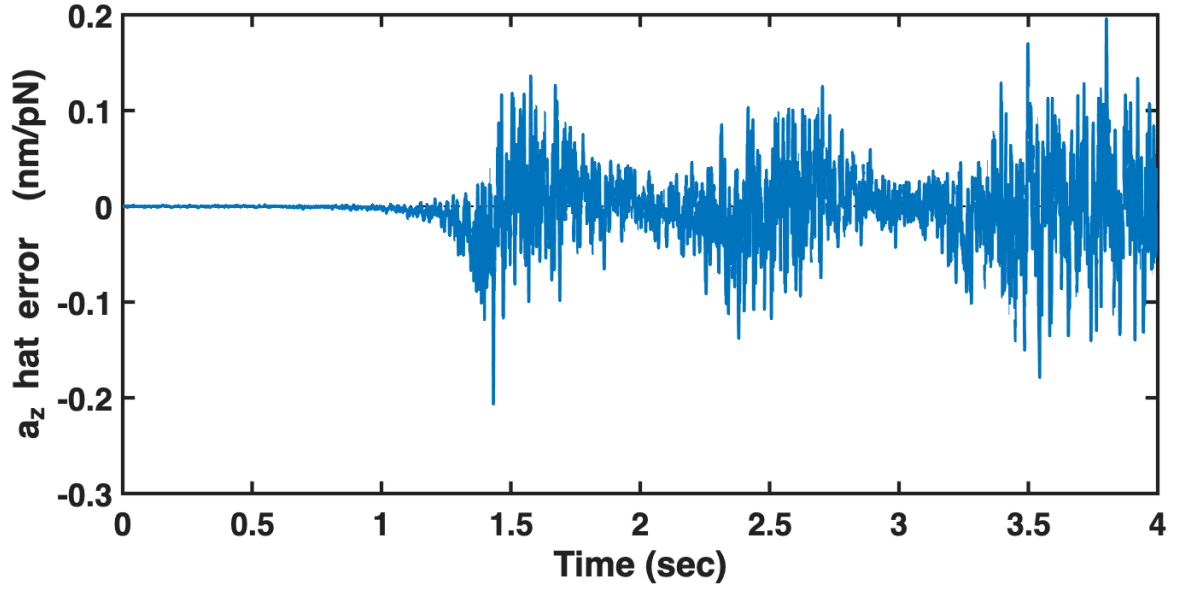
$$\text{RW: } P_a[k+1] = P_a[k] + Q_{44} \nearrow \infty \quad \text{AR(1): } P_a[k+1] = \lambda_c^2 P_a[k] + q_4 \rightarrow \frac{q_4}{1-\lambda_c^2}$$

Fix: move the reversion into the dynamics ($\mathbf{F}_e(4,4) = \lambda_c$, predict reverts to $a_{x_{det}}$), keep only the white driver in Q_{44} :

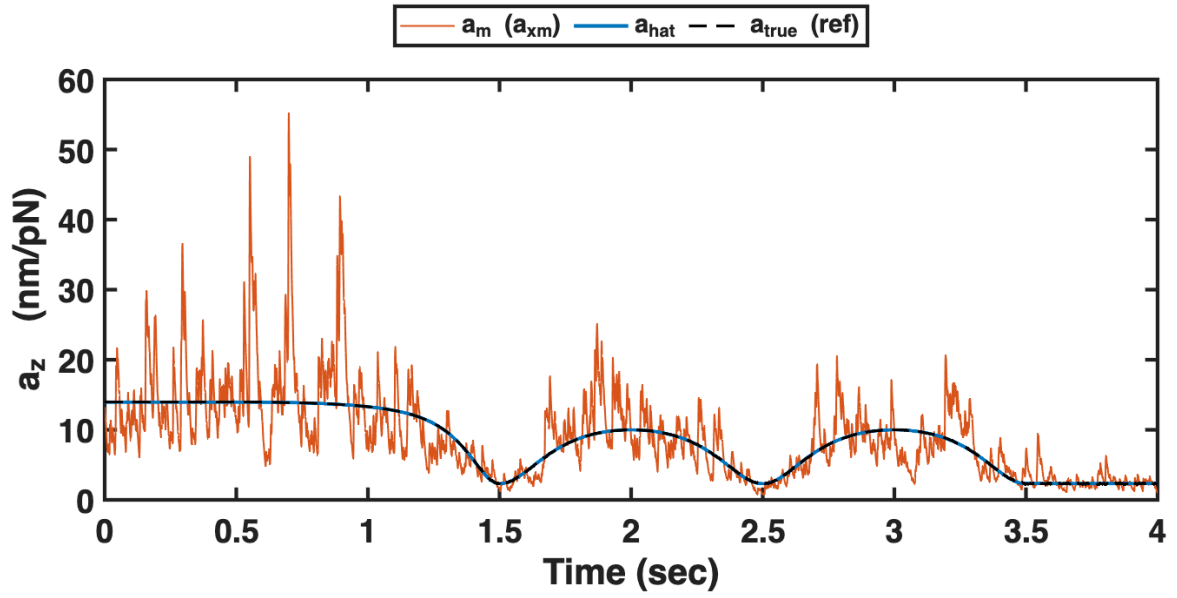
$$\underbrace{\text{Var}(\Delta a_{ram}) = \frac{2}{1+\lambda_c} a_x'^2 \sigma_{\delta h}^2}_{\text{RW } Q_{44}} \longrightarrow \underbrace{q_4 = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2}_{\text{AR(1) } Q_{44}}$$

Effect (1 Hz, near wall, z -axis): $P_a/\text{Var}(a_{x_{ram}}) : 19.67 \rightarrow 0.78, \quad |\hat{a}_z - a_z| : 9.4\% \rightarrow 0.4\%.$

Single-seed time series (1 Hz, near wall, z -axis)



a_z estimation error $\hat{a}_z - a_z$.



a_z : measurement a_m , EKF estimate $\hat{a}_x$, truth a_{true} .